Two-coordinate affine rotation table: report

Question and scope

Does storing an additional transformed x coordinate make radix-14 affine-layer batch evaluation faster by replacing constant field multiplications during leaf selection? The candidate stores `(x, m, y)` with `m=x+ x`. The identities `m−x= x` and `−m= ²x` follow from `²+ +1=0`; TwoX.lean checks the abstract additive-group implications with explicit hypotheses. The native overlay is hash-guarded, and independent binary replay checks the curve-specific implementation.

Result

The frozen CPU 40 panel used 16 alternating repetitions for the comparison plus adjacent A/A controls. All 96 arms and 48 pairs completed, source/binary/fixture artifacts were stable, every output was verified, and all A/A ratios passed `[0.90, 1.11]`. The independent audit is `tier1-two-x-cpu40-audit.json`; it checks every arm against the manifest, fixture and hashes.

Metric	Affine-layer reference	Two-x candidate
Mean online time, 12,288 outputs	56.458543 ms	54.416974 ms
Mean time per scalar	4.594608 µs	4.428465 µs
Mean table preparation	4,196.115 ms	4,230.430 ms
Retained bytes	26,035,988	37,192,948

The predeclared median paired reference/candidate ratio is **1.037997×**, with paired bootstrap 95% interval [**1.025659, 1.053525**] over 16 comparison repetitions. This qualifies a batch-throughput comparison under the frozen panel on the observed CPU 40 instance. It does not establish a one-scalar latency gain; physical-host exclusivity is unverified. The candidate uses 11.16 MiB more retained table memory and table preparation is 34.315 ms longer in the means.

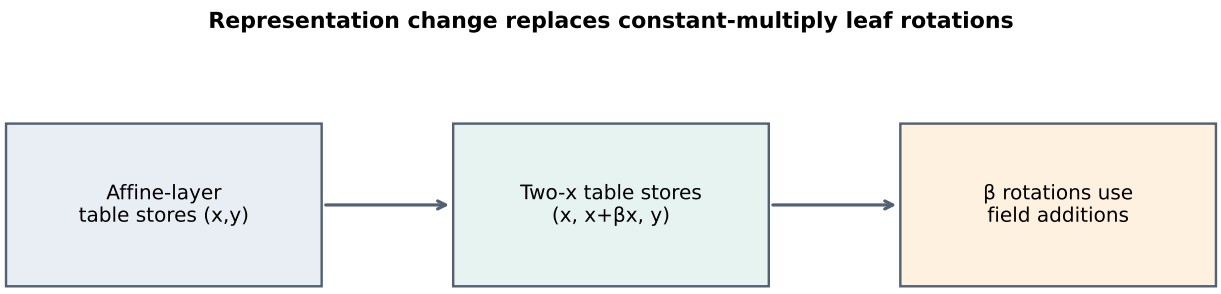
The generic runner's `paired_speedup=1.006040` includes the two A/A case medians and is not the comparison estimate. The case-specific audit applies A/A only as noise gates. The overlay's source hashes, native release binary, and retained first failed compile are described in the README and provenance archive.

Protocol and figures

The panel manifest and rules freeze the 12,288-output fixture, CPU affinity, case order, repetition count and A/A range. Online time excludes table preparation, which is measured separately; every result is verified after timing. The report figures are generated from the committed audit JSON by `visuals.py`, with their SVG and PDF sources committed here. Rebuild with:

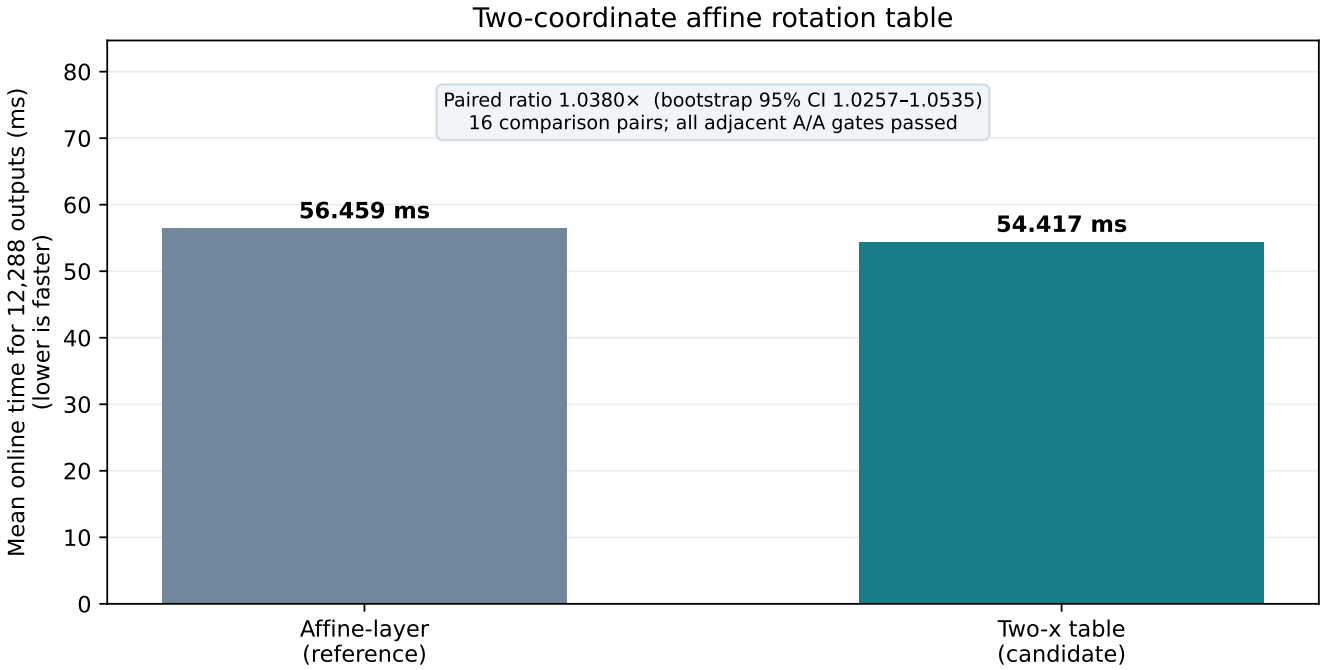
```
python3 experiments/prime-j0-affine-two-x-20261015/visuals.py
pandoc experiments/prime-j0-affine-two-x-20261015/REPORT-2026-10-19.md \
  --from=gfm --pdf-engine=xelatex -V geometry:margin=0.75in \
  -o experiments/prime-j0-affine-two-x-20261015/REPORT-2026-10-19.pdf
```

Representation diagram (SVG) · PDF



The identity follows from $\beta^2+\beta+1=0$. The panel measures prepared-table batch throughput; curve-specific correctness is checked by the native replay and every timed output is verified.

Throughput figure (SVG) · PDF



CPU 40 · frozen 12,288-scalar fixture · table preparation reported separately

Decision and limits

The evidence supports the two-x table for the specified prepared-table batch workload. Higher memory and preparation costs are material when choosing the representation. The next comparison should keep input generation, table preparation, online batch work and one-scalar cold cost distinct.